

Ni Catalyst Activity in Methane Reforming

Ni catalyst loading increased from 2 wt% to 5 wt% across batch experiments. The reactor temperature rose from 650 C to 720 C during the first hour. Grid power supply remained stable at 220 V throughout the baseline period.

Ni catalyst loading increased from 2 wt% to 5 wt% across batch experiments. The reactor temperature rose from 650 C to 720 C during the first hour. Grid power supply remained stable at 220 V throughout the baseline period.

Reaction Rate and Yield

Reaction rate rose sharply after catalyst activation. CH₄ conversion increased from 45 percent to 78 percent. Secondary products decreased when pressure was reduced.

Reaction rate rose sharply after catalyst activation. CH₄ conversion increased from 45 percent to 78 percent. Secondary products decreased when pressure was reduced.

Heat Transfer and Efficiency

Heat input to the reformer increased due to exothermic side reactions. Thermal efficiency improved when insulation was upgraded. Coolant flow rate decreased slightly at peak load.

Heat input to the reformer increased due to exothermic side reactions. Thermal efficiency improved when insulation was upgraded. Coolant flow rate decreased slightly at peak load.

Conclusions

Overall yield increased under optimized Ni catalyst conditions. The strongest causal chain links catalyst activation to reaction rate and final yield. Further work should verify upstream feed composition effects.

Overall yield increased under optimized Ni catalyst conditions. The strongest causal chain links catalyst activation to reaction rate and final yield. Further work should verify upstream feed composition effects.